

Tomas Petricek

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Academic education

- **Habilitation, Faculty of Mathematics and Physics, Charles University**, 2026
Thesis [Simple Programming Tools for Data Exploration](#) uses programming language theory research methods to develop novel data science tools. The thesis summarizes research on F# type providers, data journalism and programming tools done in [Microsoft Research](#), [The Alan Turing Institute](#) and at the [University of Kent](#).
- **PhD, Computer Laboratory, University of Cambridge**, 2011 – 2016
Thesis [Context-aware Programming Languages](#) develops coeffects, a theory for tracking information about environment in which programs are executed. I also devised a novel way of presenting the results in the form of widely read interactive essay (tomasp.net/coeffects). Supervised by [Prof. Alan Mycroft](#).
- **BSc and MSc, Computer Science, Charles University, Prague**, 2004 – 2010
MSc completed with distinction. Final thesis [Reactive Programming with Events](#), supervised by Dr Don Syme, developed abstractions for reactive and concurrent programming that were presented in workshop papers.

Professional experience

- **Associate Professor, Charles University, Prague**, 2026 – present
- **Assistant Professor, Charles University, Prague**, 2022 – 2026
- **Lecturer, University of Kent**, 2018 – 2022
- **Collaborating Fellow and Visiting Researcher, The Alan Turing Institute**, 2016 – 2020
- **Post-doctoral Researcher and Contractor, Microsoft Research Cambridge**, 2014 – 2016

Research Funding

- **PRIMUS Research Programme, Charles University**, Award value €610,000 (2024 – 2028)
- **Fellowship, ACM History Committee**, Award value \$4000 (2018)
- **Dstl & GCHQ grants, The Alan Turing Institute**, Award value £420,000 (2017 – 2020)
- **Innovation Fund, Google Digital News Initiative**, Award value €50,000 (2016)

Professional Membership

- **IFIP TC2 Working Group 2.16 Language Design** (2024 – present). Member of an invitation-based group of 40 leading programming language experts that meets twice a year for an intensive week-long meeting.

Awards

- **Reviewers' Choice Award**, «Programming» 2023. Awarded for a paper on programming systems.
- **Reviewers' Choice Award**, «Programming» 2020. Awarded for a paper on data exploration tools.
- **ACM SIGPLAN Research Highlight**. My first-author paper was chosen as an ACM SIGPLAN Research Highlight, making it one of the three best programming language papers of the past year.
- **Editors' Choice Award**, «Programming» 2018. Awarded for a paper on the history of monads.
- **Reviewers' Choice Award**, «Programming» 2017. Awarded for a paper on the history of errors.
- **Distinguished Paper Award**, PLDI 2016. Awarded for a paper on types for semi-structured data

Keynotes and Invited Talks

- **Undone Science in Computer Science**, Luxembourg (March 2026). [Keynote](#): Undone Ideas on Programming
- **PLISS Summer School**, Bertinoro (June 2025). [Invited Talk](#): Programming systems deserve a theory too!
- **Workshop of The Psychology of Programming Interest Group (PPIG)**, Belgrade, Serbia (September 2025). [Keynote](#): Critical Architecture / Software Theory
- **Data Science in F#**, Berlin (September 2023). [Keynote](#): Designing composable visualizations
- **ASL/APA Meeting**, San Francisco (April 2023). [Invited Talk](#): Cultures of programming
- **Huawei Global Technology Summit**, Edinburgh (July 2022). [Invited Talk](#): Rethinking data exploration tools
- **CodeMesh**, Virtual (November 2020). [Keynote](#): Cultures of programming
- **ScalaDays**, Berlin (May 2018). [Keynote](#): Functionalist programming language design
- **Data Science Summit**, Cambridge (June 2017). [Keynote](#): The Gamma: Democratizing data science
- **ManLang Conference**, Prague (Sept 2017). [Keynote](#): Language Challenges of Targeting Multiple Runtimes

Recent publications

For historical reasons, highly-selective conference publications are preferred over journal papers in computer science. I am the author of 20 papers in highly-selective venues (including POPL, PLDI, ICFP, ICALP, ECOOP) and 26 other peer-reviewed papers. I have an h-index of 16 (7 in WoS) and my work has 1080+ citations (233 in WoS).

- **Tomas Petricek**. [Cultures of Programming: The Development of Programming Concepts and Methodologies](#). 420 pages, Cambridge University Press, ISBN 9781009492379, 2026
- **Tomas Petricek**, Jonathan Edwards. [Denicek: Computational Substrate for Document-Oriented End-User Programming](#). Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology (UIST '25), no. 32, pp. 1-19, [10.1145/3746059.3747646](#), 2025
- **Tomas Petricek**, Joel Jakubovic. [On the Limits of Making Programming Easy](#). Languages, Compilers, Analysis-From Beautiful Theory to Useful Practice: Essays Dedicated to Alan Mycroft on the Occasion of His Retirement, LNCS, vol. 15500, Springer, [10.1007/978-3-032-08187-2_11](#), 2025
- Peter Taylor-Gooby, **Tomas Petricek**, Jack Cunliffe. [Covid-19, Charitable Giving and Collectivism: A Data-Harvesting Approach](#). Journal of Social Policy, vol. 52, issue 3, pp. 473-494, Cambridge University Press (IF: 1.9), [10.1017/S0047279421000714](#), 2023
- Joel Jakubovic, Jonathan Edwards and **Tomas Petricek**. [Technical Dimensions of Programming Systems](#). The Art, Science, and Engineering of Programming, vol. 7, issue 3, no. 13, [10.22152/programming-journal.org/2023/7/13](#), 2023
- **Tomas Petricek**, Gerrit J. J. van Den Burg, Alfredo Nazábal, Taha Ceritli, Ernesto Jiménez-Ruiz and Christopher K. I. Williams. [AI Assistants: A Framework for Semi-Automated Data Wrangling](#). IEEE Transactions on Knowledge and Data Engineering, vol. 35, issue 9, pp. 9295-9306 (IF: 8.9), [10.1109/TKDE.2022.3222538](#), 2023
- **Tomas Petricek**. [The Gamma: Programmatic Data Exploration for Non-programmers](#). VL/HCC '22: Proceedings of the IEEE Symposium on Visual Languages and Human-Centric Computing, [10.1109/VL/HCC53370.2022.9833134](#), 2022
- Roly Perera, Minh Nguyen, **Tomas Petricek** and Meng Wang. [Linked Visualisations via Galois Dependencies](#). Proceedings of the ACM on Programming Languages, vol. 6 (POPL), pp. 1-29 (IF: 1.8), [10.1145/3498668](#), 2022
- **Tomas Petricek**. [Composable Data Visualizations](#). Journal of Functional Programming, vol. 31, e. 13, Cambridge University Press (IF: 1.1), [10.1017/S0956796821000046](#), 2021